

AIML401: Assignment 3 Task 2: Language Detection

Peyton Mou (2212984)

October 15, 2024

This assignment task has compared two kinds of language detection libraries (langdetect and lingua) under scenarios of normal test and limit test, a brief discussion about their performances and limitations are presented as below:

1. Langdetect

1.1 Normal Test Performance

$$\begin{bmatrix} 86 & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 83 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 85 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 2 & 88 \end{bmatrix}$$

Above confusion matrix shows that langdetect can correctly identify the majority of sentences, with minor confusion for one instance of Italian being misidentified as Spanish. Besides, the F1-Score of 0.3920 indicates medium-level performance, showing that langdetect struggles to consistently and accurately classify languages of all sentences.

1.2 Limit Test Performance

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 2 & 0 & 0 & 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

The confusion matrix shows that langdetect fails completely in the limit test, where sentences are designed to become confusing and challenging. And the F1-Score of 0.0 means that there is no correct prediction made by this tool in the limit test scenario.

1.3 Limitations

Langdetect performs decently on straightforward and normal sentences but breaks down under confusing or mixed language test dataset. It is particularly vulnerable to mixed languages or context-switching within a sentence, and struggles with non-European languages like Chinese.

2. Lingua:

2.1 Normal Test Performance

$$\begin{bmatrix} 89 & 0 & 0 & 0 \\ 0 & 85 & 0 & 0 \\ 0 & 0 & 90 & 0 \\ 1 & 0 & 0 & 89 \end{bmatrix}$$

Above confusion matrix shows that lingua performs excellently with only one sentence error across totally 354 sentences of 4 kinds of languages. Moreover, with a very high F1-Score of 0.9972, lingua demonstrates nearly perfect classification accuracy for standard and normal sentences.

2.2 Limit Test Performance

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Above confusion matrix shows that although lingua outperforms compared with langdetect, it is still struggling with mixed-language or confusing datasets, especially when similar languages (e.g., Spanish and Italian) are involved, and the F1-Score of 0.6416 indicates that it can predict the language of confusing sentences to certain extent with some errors.

2.3 Limitations

Lingua outperforms extraordinarily better than langdetect, under both challenging limit test and normal test conditions. However, it still faces limitations when distinguishing between languages with similar structures (e.g., Italian and Spanish). Similar as langdetect, it is struggling when handling with sentences that mix different

languages (e.g., English and French) and might fail to predict the accurate language of the sentence.

3 Conclusion

In a conclusion, langdetect is effective for normal and unchallenging tasks with well-separated sentences in different languages, but it is struggling when handle with mixed or ambiguous sentences, and the accuracy rate is lower compared with lingua. On the contrary, lingua has done a more excellent job compared with langdetect, especially under normal test scenario, and also can accurately predict the majority of challenging and mixed sentences.